

Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

📞 +49 176 62145797 • ✉ a01750283@tec.mx • [LinkedIn](#)

Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience independently owning a data-automation project end-to-end at Deutsche Bundesbank (AnaCredit): from identifying a manual bottleneck to building and deploying the automated solution. Comfortable driving projects from idea to implementation with a high degree of autonomy, and motivated by AI agent development and low-code/no-code platforms.

Core Competencies

- **End-to-End Ownership:** Independently identified a manual bottleneck, then designed, built, and deployed the automated solution
- **Agentic AI & AI Agent Development:** Regular use of Claude Code as an AI coding agent; strong interest in building AI agents on low-code/no-code platforms
- **Programming:** Python (Pandas, NumPy, Scikit-Learn), SQL; building and extending data interfaces and pipelines
- **Data Quality & Automation:** Automated migration tests (row counts, referential integrity, aggregate consistency), deviation and anomaly detection, rule-based validation
- **Data Visualization & Analytics:** Dashboards and structured reporting (Tableau, Matplotlib, Seaborn); quick to onboard onto new platforms such as Palantir Foundry
- **Collaboration & Documentation:** Stakeholder-facing requirement translation, Jira, Git, structured technical documentation

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern 04/2026–07/2026
 - Independently identified a recurring manual bottleneck and drove the solution from idea to implementation: an automation pipeline generating Jira tickets and emails directly from detected data errors
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to verify identical business logic before roll-out
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reporting data, then visualized findings for stakeholder review
 - Significantly reduced response time to data quality issues through end-to-end ownership, from detection to resolution

Projects

- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks
 - Built predictive models (Random Forest, Neural Networks) and K-Means clustering to segment high-risk groups, directly analogous to clustering structured records into meaningful categories

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; built using **Claude Code** as an AI coding assistant to accelerate implementation
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python
 - Contributed to a scientific publication analyzing security parameters and runtime behavior across lattice dimensions

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, NLP, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026
- **Tecnológico de Monterrey** **Mexico**
Abitur 2019–2022
Grade: 1.0. National Physics Olympiad 2020; Honorable Mention, Physics Olympiad 2021.

Languages

- Spanish (native), English (C1, TOEFL iBT 105/120), German (C1).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.